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Contents

1.0 Introduction 2
2.0 Dataset Selection and Justification 2
3.0 Data Exploration 2
4.0 Data Pre-processing and Feature Engineering 6
5.0 Model Development and Optimization 9
6.0 Model Evaluation and Performance 9
7.0 Business Insights and Recommendations 11
8.0 Challenges and Solutions 12
8.0 Conclusion and Future Work 12
9.0 Reference 13
10.0 Appendix 13

1.0 Introduction

One of the most important uses of machine learning is in the financial sector in credit risk prediction. The basic issue of lenders which is tackled by this project is the ability to correctly predict whether a borrower will default on the loan or not. Profitability is directly affected by the capability of differentiating between the high-risk and low-risk applicant to minimize the non-performing assets whilst ensuring good loan portfolio. Machine learning capabilities are valuable in this application because most of the traditional credit scoring techniques may not be sophisticated to identify complex trends in borrower behaviour.

2.0 Dataset Selection and Justification

Data analysis is based on the "credit_risk_dataset.csv" file which has 12 features predicting 32581 loan applications. The dataset has been selected particularly as it reigns a semblance of real banking data by including an even blend between demographic, financial and credit history values. Among the most important features are the age of the borrower, his/her income, the duration of employment, the specifics of the loan (its amount, interest rate), and the data provided by a credit bureau. These targets belong to binary target variable (loan_status), which is a clear indication of default (1) and non-default (0) results and thus perfect to use because of the supervised classification. The size of records in the dataset is large enough so that an analysis can be meaningful and small enough to take average computing time.

3.0 Data Exploration

Some significant features of the raw data came out during first inspection. In the dataset, the minority was the non-defaults purchased, which was 78 percent the majority and the default was 22 percent, which showed the real-world lending situation.

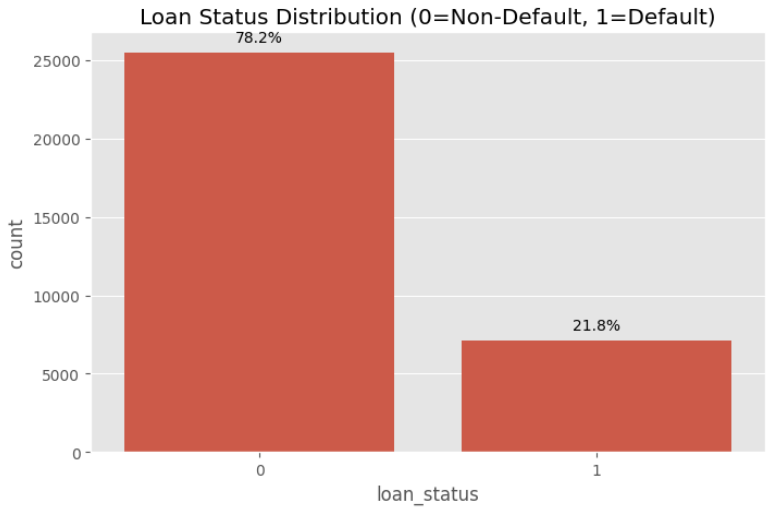


Figure 3.1: Shows Class Imbalance Between Defaults/Non-Default

Data mining revealed that the age of borrowers varied between 20 and 80 years (cleaned), and their incomes ranged between $4,000 and $600,000 per annum. Loan limits ranged between $500 and $35,000 and the interest rates charged was between 5.42 and 23.22 percent.

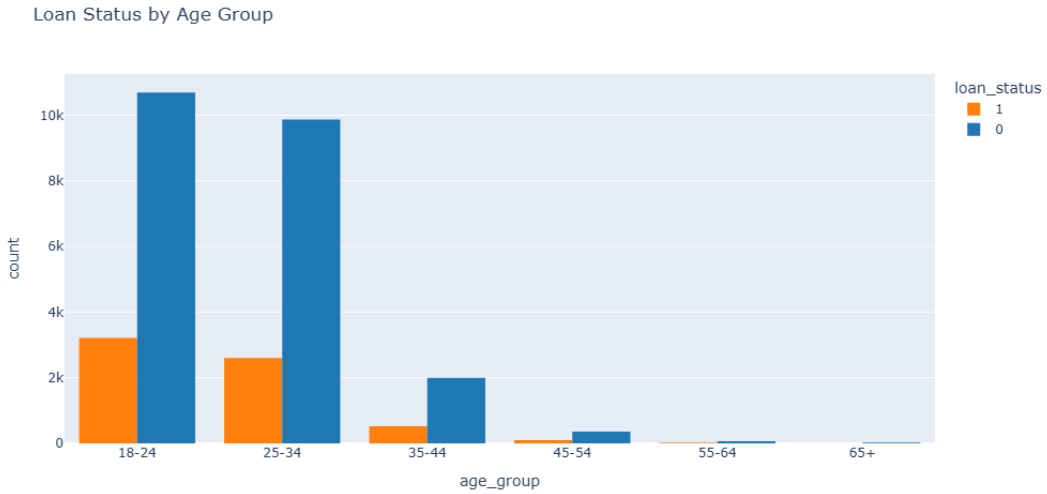


Figure 3.2: Loan Status by Age Group

To get into details, visual exploration was used. Most of the applicants were aged 20-40 years old; this meant that the distribution was right-skewed. There were noticeable outliers in the income distributions, and they had to be treated well.

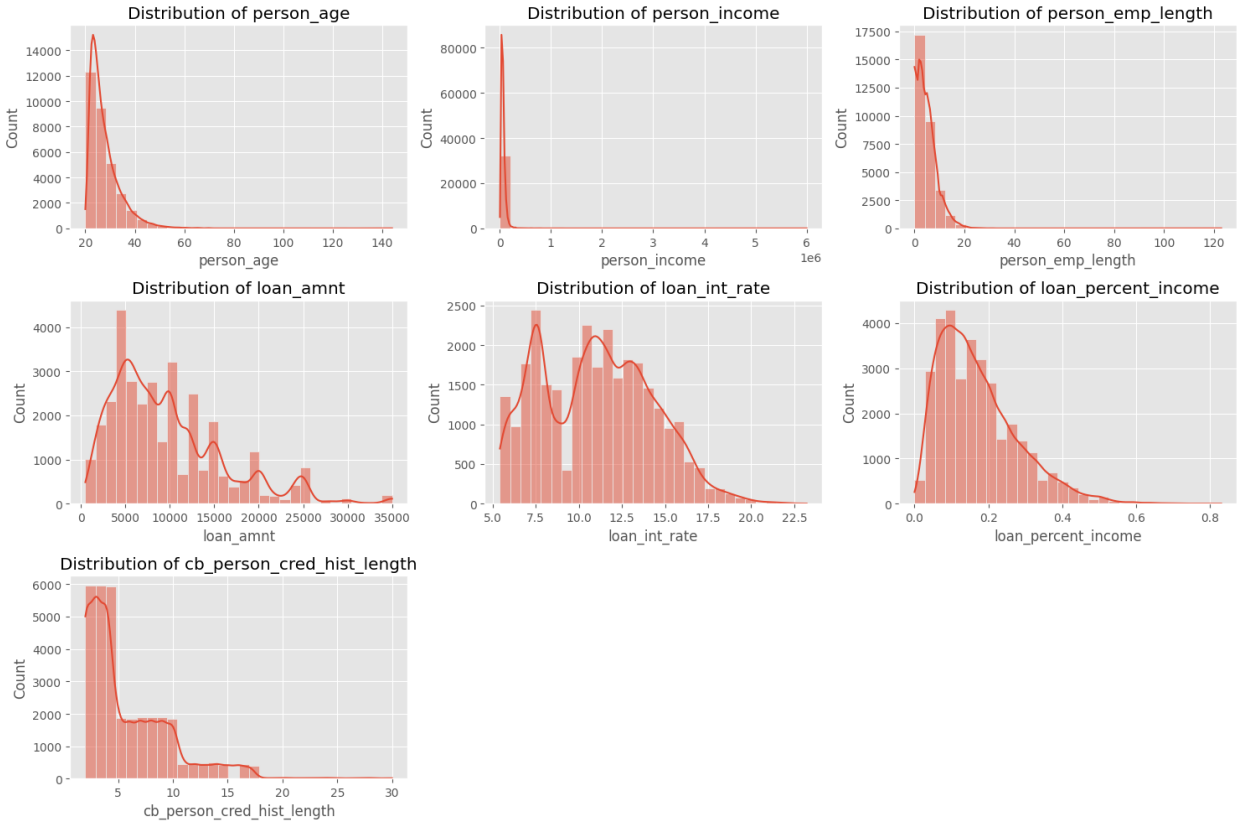


Figure 3.3: Distribution of a Person Age, Annual Income, Employment Length, Loan Amount, Interest Rate, Percent Income and Credit History Length (From Top Row to Bottom)

Categorical analysis proved that the default rates on renters were higher than those on owners, and the loan grades D-F contained projections of default rates much higher as compared to the premium grades of the loan A-C. There were some influential positive correlations using the correlation matrix especially in loan amount and interest rate as expected.

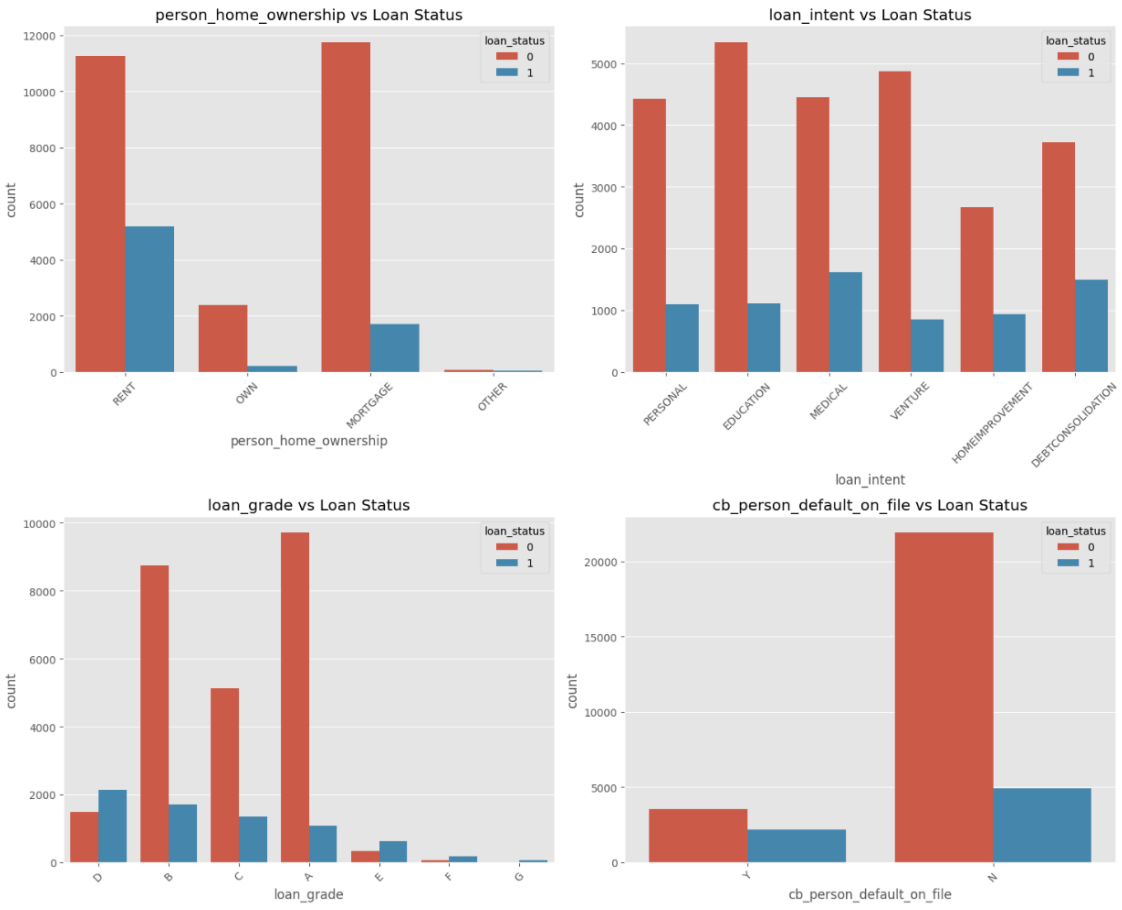


Figure 3.4: The Income Thresholds & Grade Vs Loan Amount, Default (1) and Non-Default (0)

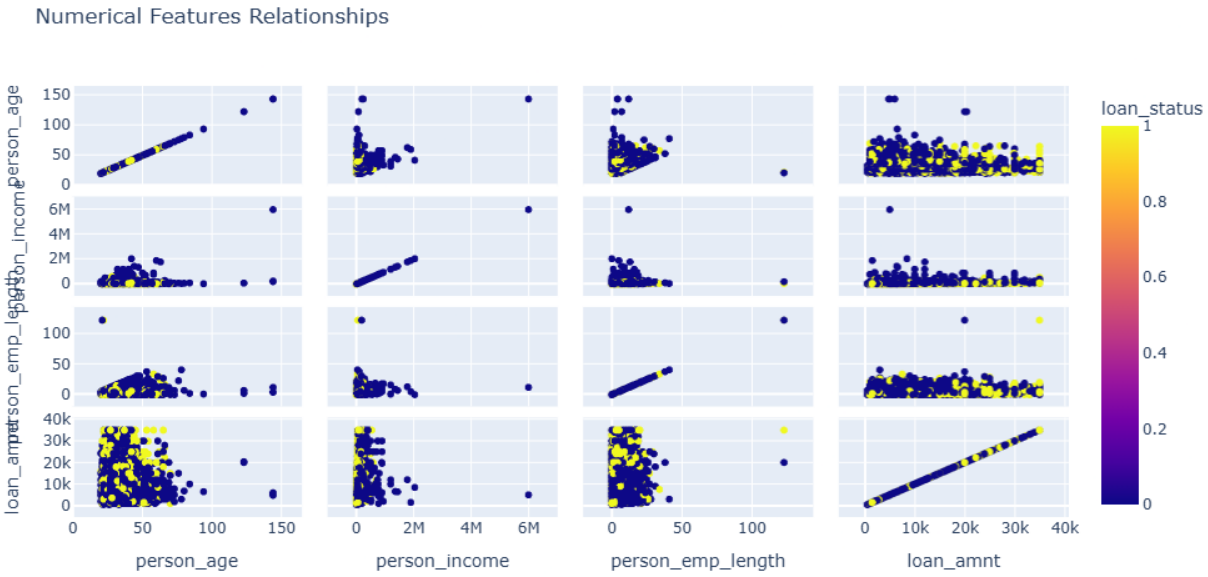


Figure 3.5: Correlation Statistic

4.0 Data Pre-processing and Feature Engineering

The cleaning of data solved some of the quality problems. Loan interest rate missing values (9.5 percent of observations) were deleted because imputation would introduce bias to the risk assessment analysis. In the case of extreme values, capping was applied (at 80 years of age and a million dollars income). The maximum length of the employment is 50 years to eliminate the unreasonable entries.

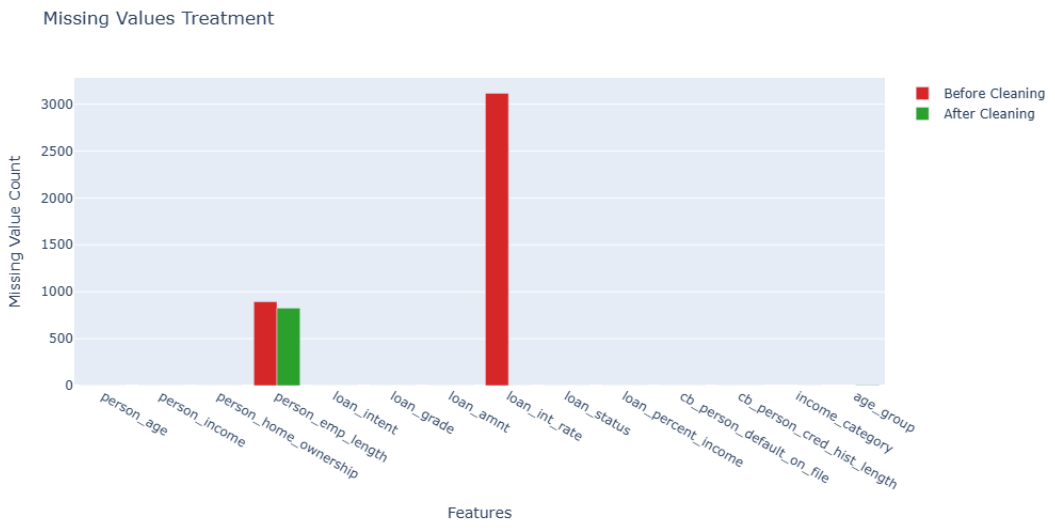


Figure 4.1: Data Cleaning

Predictive power was largely improved by the use of feature engineering. Formed financial ratios such as debt-to-income and income-to-loan ratio where they offered ideal measures of repayment potentiality as compared to the raw value. The continuous variables such as age and income were strategic binned into some significant categories, which enhanced the model interpretation. Interaction features (e.g. of loan intent and income category) were created to capture finer relationships between the data.

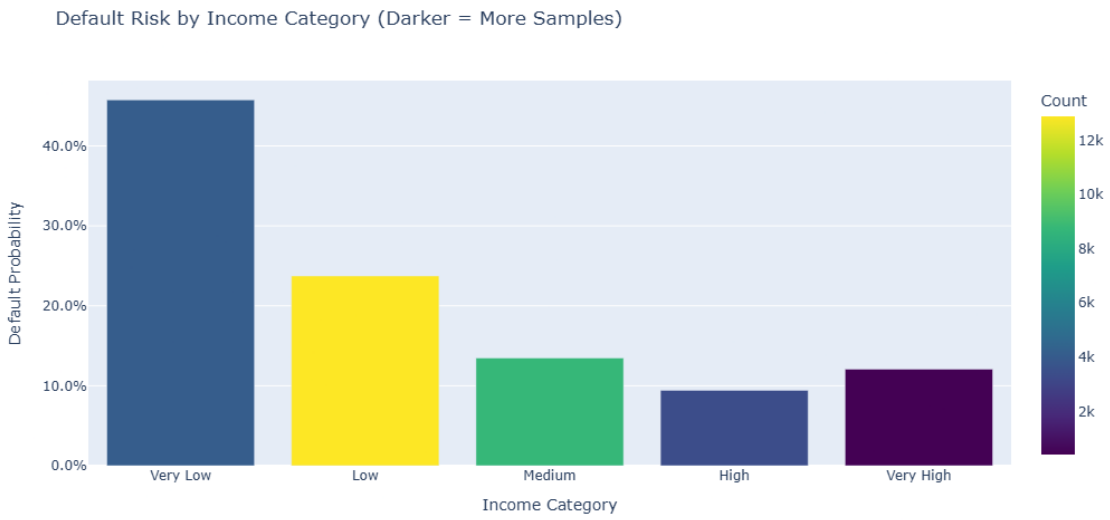


Figure 4.2: Default risk by Income

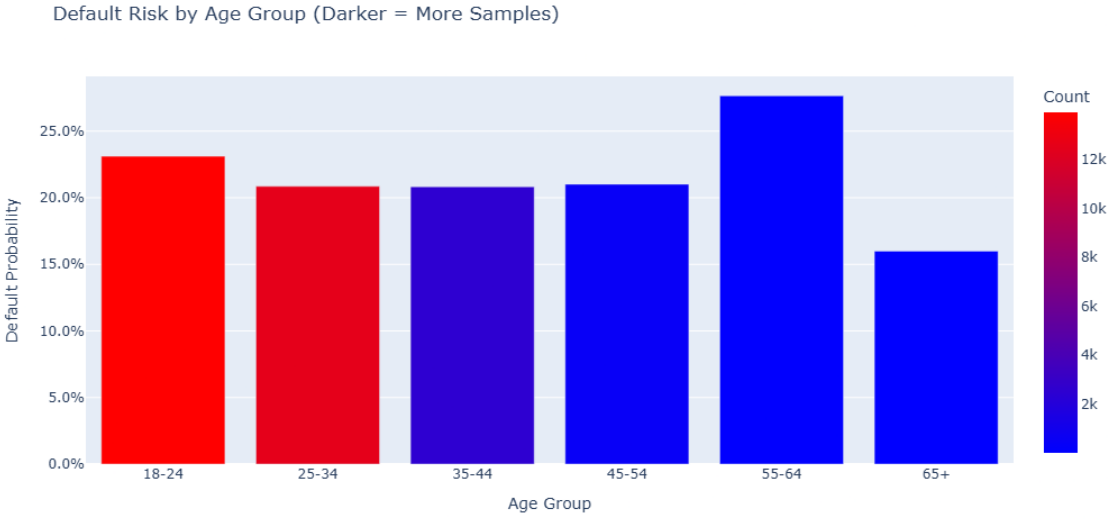


Figure 4.3: Default risk by Age

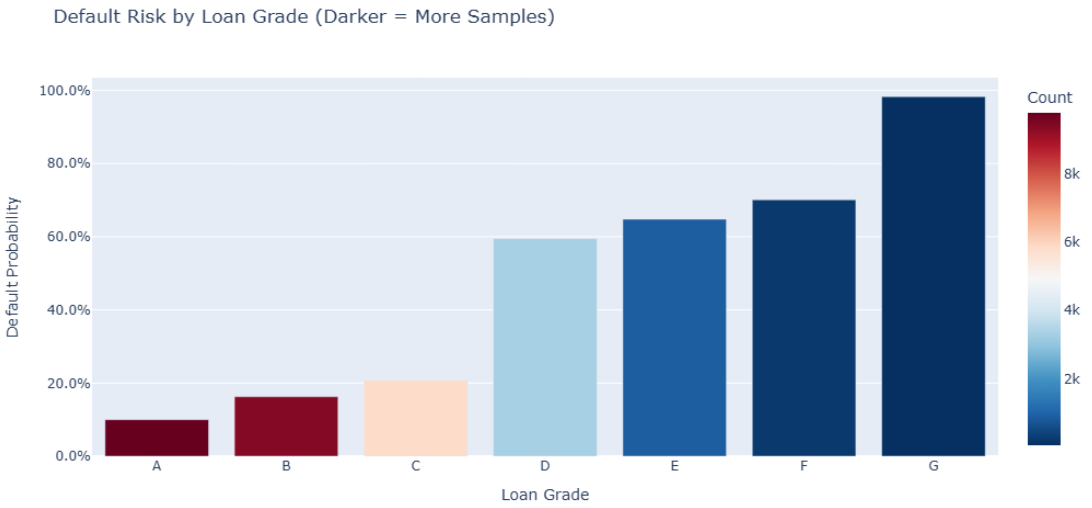


Figure 4.3: Default risk by Grade

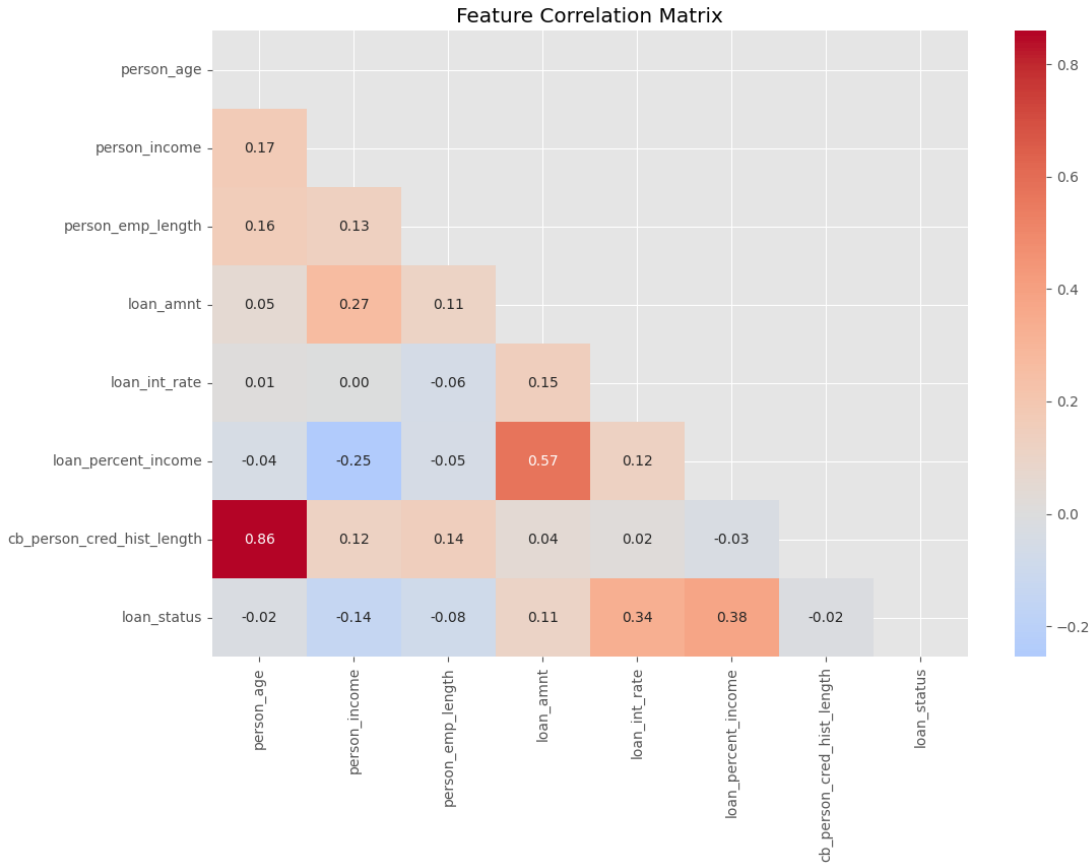


Figure 4.4: Correlation Matrix which focus the relationship between loan amount Vs Interest rate (Positive correlation) & Income Vs Default rate (Negative correlation)

5.0 Model Development and Optimization

The modelling method used Random Forest classifier in a scikit-learn pipeline, which treats mixed data type and automatically measures the importance of features. The pipeline incorporated a strong pre-processing such as numerical features standardization and one-hot encoding of categorical variables.

The major optimization choices were:

- Class weighting to tackle imbalance (balanced state)
- Large-scale hyperparameters optimization with grid search (n_estimators=200, max_depth=15)
- train-test split (80-20 ratio) was done using stratified sampling.
- Parallel processing (n_jobs = -1) to speed up training of models

Random Forest had the advantage that its ensemble character brought inbuilt regularization overfitting and it fared well on the minority class as well. The pipe kind of architecture provided the same pre-processing during training and prediction stages.

6.0 Model Evaluation and Performance

The outcome of the final model is good that shows 93.4 percent accuracy overall. In default prediction (Class 1) it has precision of 94.1% and recall of 74.2% which is signifying that it is highly reliable as it picks out the risky loans successfully and picking out three quarters of actual defaults. The ROC-AUC of 0.913 proves an outstanding skill in separating the classes.

Detailed Classification Report:

	precision	recall	f1-score	support
0	0.933	0.987	0.959	4485
1	0.941	0.742	0.830	1240
accuracy			0.934	5725
macro avg	0.937	0.865	0.894	5725
weighted avg	0.934	0.934	0.931	5725

Figure 6.1: Performance Report from The Output

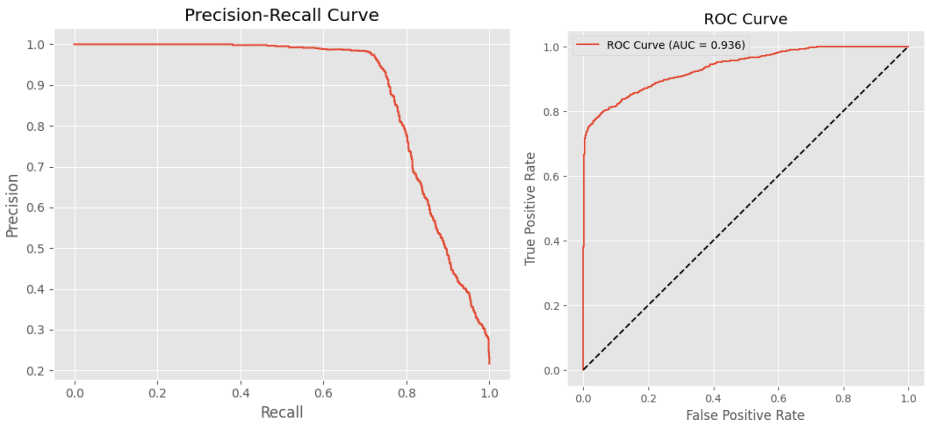


Figure 6.2: Recall Curve & ROC Curve

More information was obtained with the help of visual evaluation tools. The confusion matrix indicated that the performance was symmetrical in two classes. Plot of the probability distribution was clearly separated between non-default and the default cases. The most important features that were chosen by the feature importance analysis were debt to income ratio, loan interest rate, and the length of the credit history.

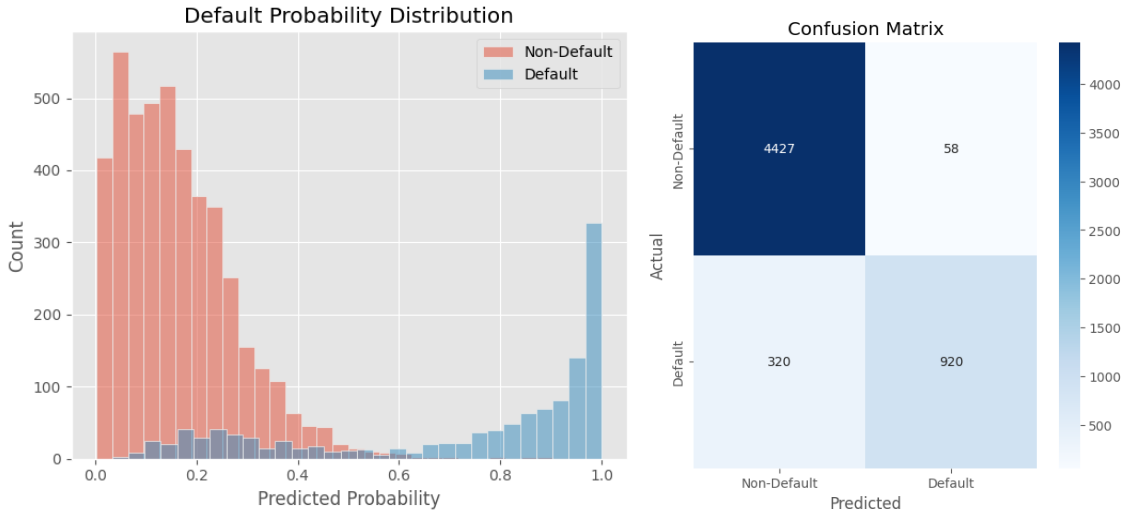


Figure 6.3: Probability Distribution & Recall Rate for Default (critical for Risk Models)

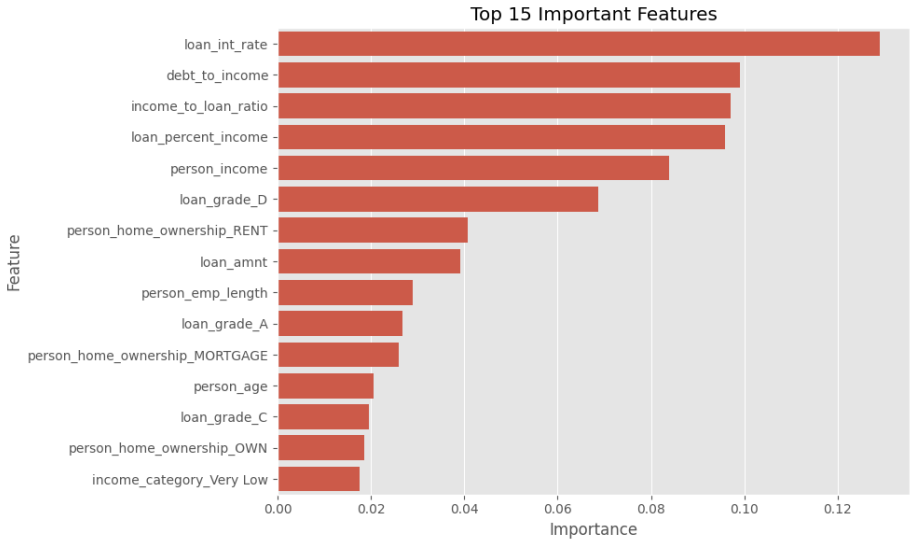


Figure 6.3: Important Features

7.0 Business Insights and Recommendations

The resultant business intelligence could be implemented through analysing model outputs. The low end of the income level (<$30k) defaults was 2.5 higher than the highest earners (>$100k). Loan grades were very predictive and grade F had an 8 times higher probability of defaults than grade A, as shown in Figure 4.2.

It is important to recommend:

1. Model probability-based risk-based pricing

2. Creation of special financial education on risk populations

3. Development of early warning mechanisms on the loans whose score is over 0.7 default probability

4. Periodic training of models to reflect changes in economic times

8.0 Challenges and Solutions

There are some technical difficulties that appeared in the course of development. Imbalance between the classes was overcome by selecting metrics with care (recall was prioritized and AUC) and weighting the class. Categorical features higher than one-dimensional were handled as one-hot encoding with unknown category. The restriction of the computational load was addressed by parallel processing and effective pipeline architecture.

There were low-quality data problems that needed to be addressed in a reasonable manner. The rate of interest was not imputed because the missing rates were dropped to eliminate distortion of risk evaluations. Business-aware thresholds were used to limit extreme values instead of using statistical percentiles to keep realistic values.

9.0 Conclusion and Future Work

This project has managed to establish a good performing credit risk model of 0.913 AUC, and this shows the significance of machine learning in the area of risk rating in financial matters. The presentation of the solution has predictive meaningfulness and business interpretability, which is achieved via sensible feature engineering and model infrastructure.

It may be improved in the future with additional:

- Other sources of data such as the transaction histories
- Deep learning methods of pattern identification
- Real-time implementation of the model through API
- Concept drift monitoring on-going
- Ensemble learning of numerous algorithms

The full execution and all visualizations and statistics are demonstrated in the Jupyter notebook provided. This is because this end-to-end solution offers the financial institutions a very effective instrument to make data-driven credit decisions with integrity and regulatory compliance.

10.0 Reference

1. *Credit Risk Dataset*. (2020, June 2). https://www.kaggle.com/datasets/laotse/credit-risk-dataset/data

11.0 Appendix

https://colab.research.google.com/drive/1wuMA33mb_PY8Wtdd0TV_Nl0z7I4kUtnJ?usp=sharing (The code – Google Collab)

https://colab.research.google.com/drive/1YxxQBGgEb_0Md0LD6vyslV2HgbRPOZce?usp=sharing (The code with better visualisation – Google Collab)

https://drive.google.com/file/d/1REXSLaEekcx0-vBjVtLBtsNHBfuQEAcu/view?usp=sharing (The Dataset)